

PHISHING AWARENESS TRAINING REPORT

CodeAlpha Cyber Security Internship

Task 2: Phishing Awareness Training

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1. Introduction

Phishing is one of the most common cybersecurity threats affecting individuals and organizations worldwide. Cybercriminals use deceptive techniques to trick users into revealing sensitive information such as usernames, passwords, banking details, and personal information. These attacks are commonly delivered through fake emails, malicious websites, text messages, or phone calls that appear to come from trusted organizations.

The purpose of this awareness training is to educate users about phishing attacks, explain how they work, identify common warning signs, and provide practical techniques to prevent becoming a victim.

2. Objectives

The objectives of this project are:

- Understand the concept of phishing attacks.
- Learn different types of phishing techniques.
- Identify fake emails and malicious websites.
- Understand social engineering tactics.

- Study a real-world phishing incident.
 - Learn cybersecurity best practices.
 - Evaluate learning through an interactive quiz.
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3. What is Phishing?

Phishing is a cyberattack in which attackers impersonate trusted organizations or individuals to deceive victims into revealing confidential information.

The primary goal of phishing attacks is to steal:

- Usernames
- Passwords
- Banking credentials
- Credit card information
- Personal data

Attackers often use fake websites and fraudulent emails that appear legitimate to gain the victim's trust.

4. Types of Phishing Attacks

Several forms of phishing attacks are commonly used by cybercriminals.

Email Phishing

The most common phishing attack where fake emails pretend to come from trusted companies or organizations.

Spear Phishing

A targeted phishing attack aimed at a specific individual, employee, or organization.

Whaling

A phishing attack targeting executives and senior management personnel.

Smishing

Phishing conducted through SMS or text messages.

Vishing

Voice phishing carried out through fraudulent phone calls.

Clone Phishing

A legitimate email is copied and modified by replacing genuine links or attachments with malicious content.

5. Recognizing Phishing Emails

Phishing emails often contain warning signs that users should recognize before responding.

Common indicators include:

- Unknown or suspicious sender
- Urgent or threatening language
- Spelling and grammar mistakes
- Suspicious hyperlinks
- Unexpected attachments
- Requests for passwords or One-Time Passwords (OTPs)

Users should always verify the sender and inspect links before clicking them.

6. Fake Websites

Cybercriminals frequently create fake websites that closely resemble legitimate websites in order to steal sensitive information.

Before entering any personal information, users should verify:

- HTTPS connection
- Correct website address (URL)
- Valid SSL certificate
- Professional website design
- Contact information
- Spelling of the domain name

Even a small difference in a website address may indicate a phishing website.

7. Social Engineering

Social engineering is the psychological manipulation of individuals to persuade them to reveal confidential information or perform actions that compromise security.

Attackers commonly exploit:

- Fear
- Urgency
- Curiosity
- Trust
- Authority

Understanding these manipulation techniques helps users recognize suspicious situations before becoming victims.

8. Real-World Example

Google Docs Phishing Attack (2017)

In 2017, cybercriminals launched a large-scale phishing campaign by sending fake Google Docs sharing invitations.

Victims received emails appearing to originate from trusted contacts. Clicking the provided link requested permission for a malicious application that gained access to users' accounts.

Millions of users were targeted before Google successfully disabled the malicious application.

Lessons Learned

- Verify the sender before opening attachments.
 - Check application permissions carefully.
 - Never authorize unknown applications.
 - Report suspicious emails immediately.
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9. Best Practices

The following cybersecurity practices help protect users from phishing attacks:

- Verify the sender before responding.
- Hover over links before clicking.
- Enable Multi-Factor Authentication (MFA).
- Use strong and unique passwords.
- Keep software and operating systems updated.
- Install antivirus software.
- Never share passwords or OTPs.

- Report suspicious emails to the appropriate authority.
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10. Interactive Quiz

Question 1: What is phishing?

- A. A firewall
 - B. A cyberattack that tricks users into revealing sensitive information ☒
 - C. An antivirus program
 - D. A VPN
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Question 2: Which of the following is a common sign of a phishing email?

- A. Official company domain
 - B. Urgent request for personal information ☒
 - C. Correct spelling and grammar
 - D. Personalized security certificate
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Question 3: Which website is more likely to be malicious?

- A. <https://www.microsoft.com>
 - B. <https://login-micr0soft.com> ☒
 - C. <https://github.com>
 - D. <https://www.google.com>
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Question 4: What should you do before clicking a link in an email?

- A. Click immediately
 - B. Verify the sender and inspect the URL ☒
 - C. Disable your antivirus
 - D. Ignore browser security warnings
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Question 5: Which security practice helps protect against phishing attacks?

- A. Sharing your OTP with trusted callers
 - B. Using the same password for every account
 - C. Enabling Multi-Factor Authentication (MFA) ☒
 - D. Ignoring software updates
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Question 6

Which of the following is an example of **Smishing**?

- A. A phishing email
 - B. A phishing text message (SMS) ☒
 - C. A fake website
 - D. A phishing phone call
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Question 7

What should you do if you receive a suspicious email requesting your password?

- A. Reply with the password
 - B. Delete the email or report it to the IT/Security team ☒
 - C. Click the link to verify the request
 - D. Forward it to friends
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Question 8

Which security feature adds an extra layer of protection to your online accounts?

- A. Browser History
- B. Multi-Factor Authentication (MFA) ☒

- C. Cookies
 - D. Incognito Mode
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Question 9

Why do attackers use **social engineering** in phishing attacks?

- A. To improve internet speed
 - B. To manipulate people into revealing confidential information 
 - C. To fix software bugs
 - D. To update operating systems
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Question 10

Which of the following is the **best practice** to protect yourself from phishing attacks?

- A. Use the same password for every account
 - B. Click every email link immediately
 - C. Verify the sender and inspect links before clicking 
 - D. Disable antivirus software
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11. Conclusion

Phishing remains one of the most widespread cybersecurity threats worldwide. Cybercriminals continuously develop new techniques to deceive users into revealing confidential information.

By understanding phishing attacks, recognizing suspicious emails and websites, practicing safe online behavior, and implementing security measures such as Multi-Factor Authentication, users can significantly reduce the risk of becoming victims.

This project successfully achieved the objectives of the CodeAlpha Cyber Security Internship Task 2 by providing practical phishing awareness training and cybersecurity best practices.

12. References

- Python Documentation – <https://www.python.org/>
- Google Safety Center – <https://safety.google/>
- Cybersecurity and Infrastructure Security Agency (CISA) – <https://www.cisa.gov/>
- Microsoft Security – <https://www.microsoft.com/security>
- Scapy Documentation – <https://scapy.readthedocs.io/>